

Space Radiation Risk Assessment for the Global Satellite Fleet

Dennies Bor (PhD Candidate)

Background

- 12,000+ satellites worth billions face radiation threats from solar energetic particles, causing single-event effects and degradation
- May 2024 "Gannon" storm: the most severe space weather event since 2003 (We had a G4 during early Nov of this year).

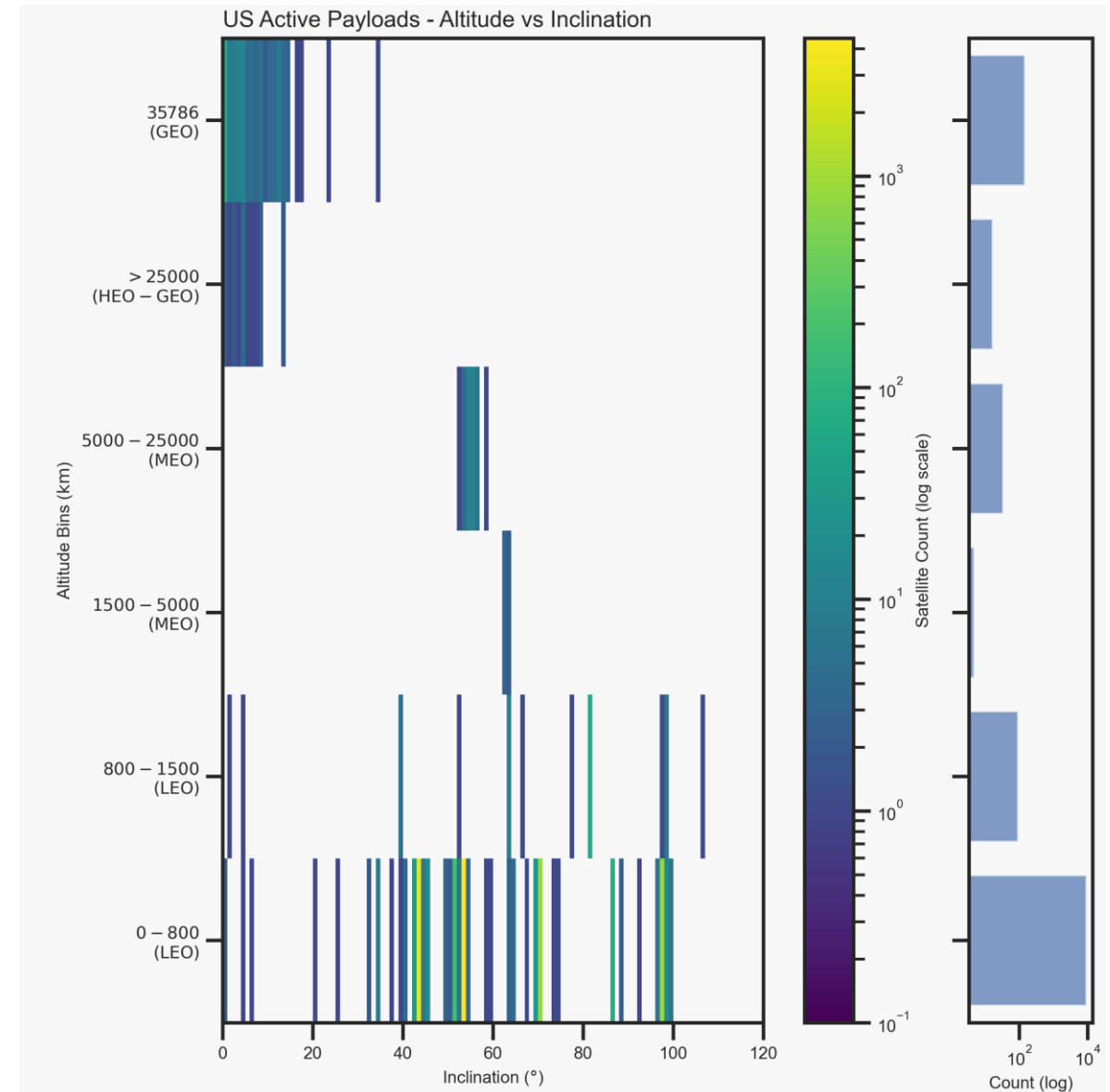
Gap: No framework linking particle flux → orbital vulnerability → economic loss with uncertainty quantification

RQ: How do we quantify radiation exposure across orbit regimes and translate physical vulnerability into probabilistic economic risk estimates?

Methodology

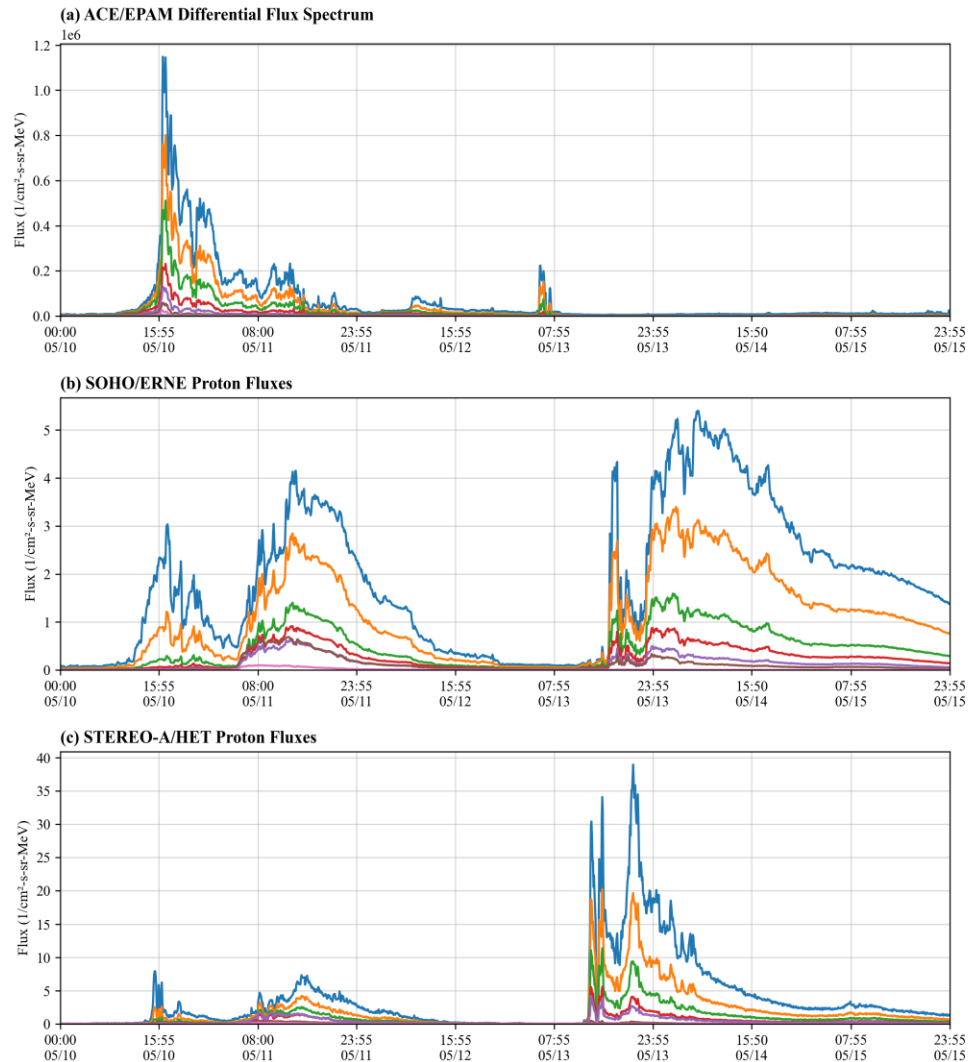
1. **Build Satellite Fleet:** Catalog ~12,000 satellites, trace orbits with SGP4, classify by regime (LEO/MEO/GEO) and mission type
2. **Model Radiation Exposure:** Measure differential flux at L1 → propagate through IGRF-13 + TSY01 fields → derive flux at each orbit
3. **Economic Valuation:** Parametric cost model based on mass, age, and mission type. Propagate direct impact through IO frameworks
4. **Vulnerability Assessment:** Apply fragility curves relating dose to failure probability

Global Satellite Fleet

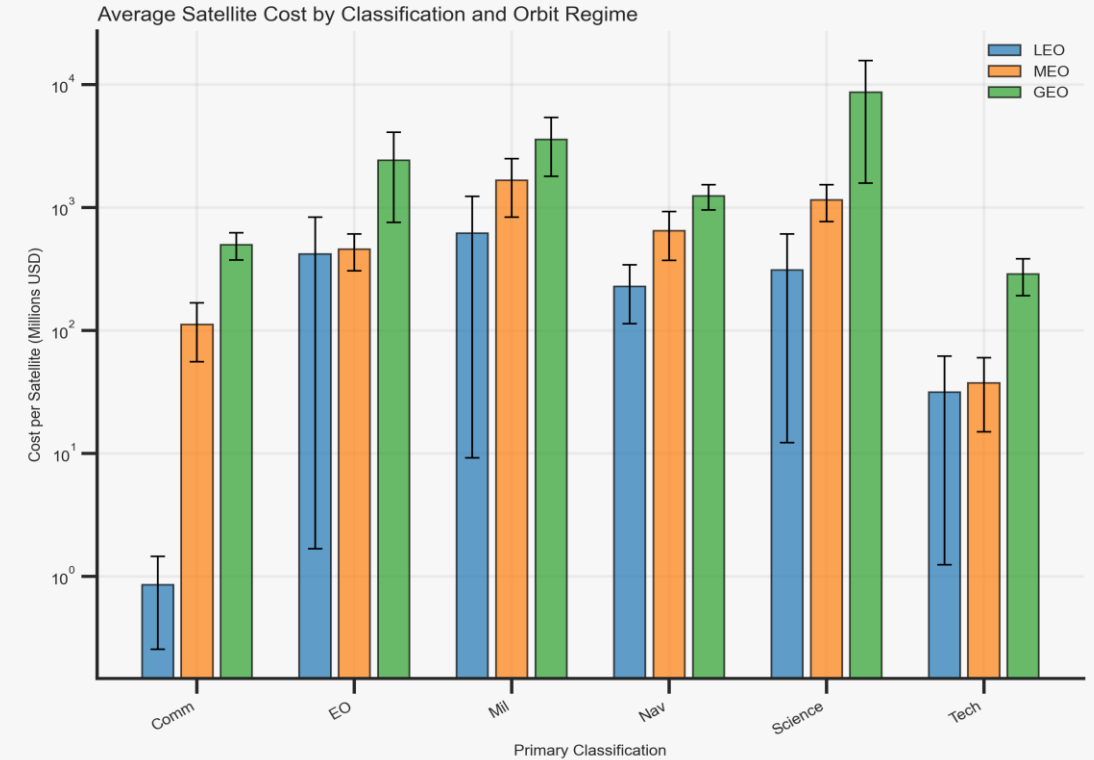


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Radiation Environment (Gannon Storm)



Economic Vulnerability



Key Findings

- GEO satellites face highest radiation exposure due to outer radiation belt location
- MEO navigation systems vulnerable during storm-time magnetospheric compression
- LEO constellations primarily vulnerable to atmospheric drag from enhanced neutral gas density; economic impacts scale with storm intensity